

Breast Cancer Image Classification Using Deep Learning

Image Processing & AI Medical Diagnosis Project

Mohamed Tarek

230104594

Mohab Badawy

230104613

Youssef Elshazly

230104603

 github.com/mohamedtarek750/Image-Processing-Project

PROBLEM STATEMENT

The Challenge

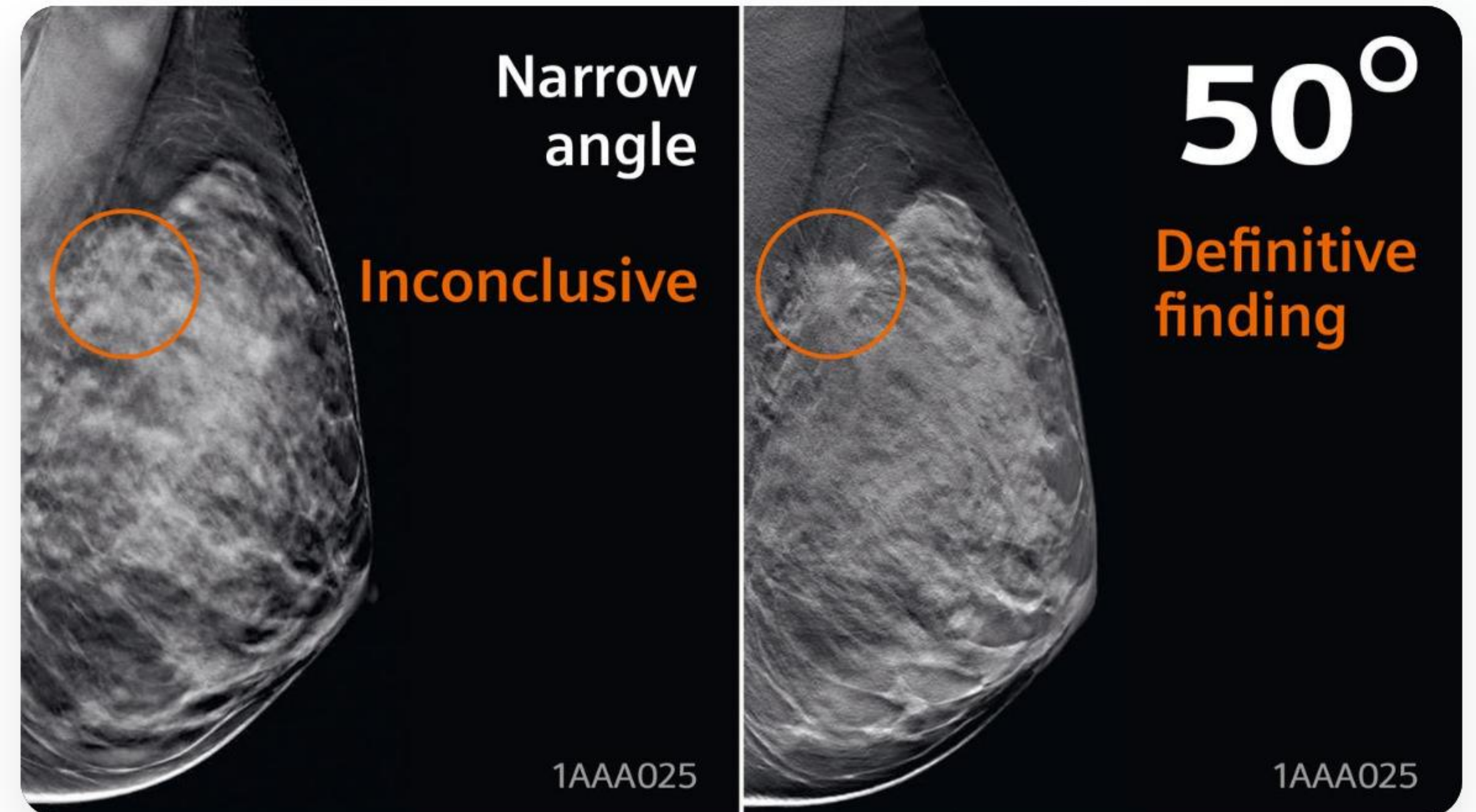
Breast cancer remains one of the leading causes of mortality globally. Early and accurate detection is critical for effective treatment. However, manual analysis of mammographic images is subject to human fatigue and inter-observer variability.

Our Objective

Develop a robust automated pipeline for **Binary Image Classification** (Cancer vs. Non-Cancer) using state-of-the-art Deep Learning models to assist radiologists in high-confidence screening.

DATASET & PREPROCESSING

-  **Binary Classification:** Focused on differentiating malignant (Cancer) from healthy tissue.
-  **Resizing:** All input images standardized to 224×224 pixels.
-  **Rescaling:** Pixel values normalized by 1/255 for optimized gradient descent.
-  **Libraries:** Powered by TensorFlow, Keras, NumPy, and Pandas.



ARCHITECTURE: SCRATCH CNN

-  Custom sequential layers designed specifically for feature extraction.
-  Deep convolutional filters to capture subtle texture variations in scans.
-  Max-pooling operations to ensure spatial invariance.
-  Fully connected dense layers for final probabilistic classification.



Baseline Model

The Scratch CNN serves as our primary baseline to evaluate how well a standard architecture can learn directly from raw medical data without pre-training.

TRANSFER LEARNING MODELS



ResNet50

Utilizes residual blocks with shortcut connections to overcome the vanishing gradient problem, enabling the training of deeper networks with high performance.



DenseNet169

Implements dense connectivity where each layer receives feature maps from all preceding layers, promoting feature reuse and reducing the number of parameters.

EVALUATION METRICS



Accuracy

Overall percentage of correct predictions.



Precision

Exactness: Ability to identify only cancer cases correctly.



Recall

Completeness: Ability to find all cancer cases (Sensitivity).



F1-Score

Harmonic mean of Precision and Recall.

*In clinical AI, **Recall** is paramount to ensure no positive case goes undetected.*

COMPARATIVE PERFORMANCE RESULTS

Model	Accuracy	Precision	Recall	F1-Score
Scratch CNN	0.9642	1.0000	0.7894	0.8823
ResNet50	0.9107	0.6956	0.8421	0.7619
DenseNet169	0.9642	0.9411	0.8421	0.8888

CONCLUSION & FUTURE WORK

96%

Top Accuracy

Key Takeaways

DenseNet169 achieved the best balance between precision and recall, making it the most suitable model for diagnostic support.

Future Work:

- Multi-class classification (Benign vs Malignant vs Normal).
- Integration of Grad-CAM for explainability.
- Expanding the dataset for greater demographic diversity.

IMAGE SOURCES



https://marketing.webassets.siemens-healthineers.com/5248af5d2b03d75d/7059ea68bc35/v/199bbe9b8744/siemens_healthineers_xp_narrow-angle-vs-50degree-01-3d-mammogram-technology_16to9.jpg

Source: www.siemens-healthineers.com